Mehul Kaushik

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Education

Manipal Institute of Technology — Manipal, India
Bachelor of Technology in Computer Science — *2023 – 2027*

Experience

Head of Cybersecurity — July 2025 – April 2026
ISTE Manipal — *Manipal, India*
- Organized CTF competitions for 150+ participants across web security, cryptography, and Linux
- Led a 20-member team to conduct cybersecurity workshops and hands-on labs
- Coordinated technical events, mentorship sessions, and practical learning initiatives

Project Intern — May 2025 – July 2025
STMicroelectronics — *Noida, India*
- Automated Python reporting workflows, reducing manual effort by 60%.
- Built modular Python scripts for CVIP PLL verification, cutting manual verification effort by 40% and accelerating regression workflows 2 times.

Projects

Merchant Co-pilot — AI Payment Assistant | *TypeScript, Node.js, Express, React, AWS Bedrock* — GitHub
- Built an AI payment assistant for 3 merchant workflows: payment queries, refunds, and settlement analysis
- Integrated AWS Bedrock with custom tools and session memory for contextual payment assistance
- Built a webhook-based fraud pipeline to score payment-event anomalies and trigger real-time alerts
- Secured webhook handling with HMAC signature verification and REST APIs

RAG-Powered Document Q&A System | *Node.js, MongoDB, Redis, Vector Search* — GitHub
- Enabled users to upload and query PDFs through a conversational LLM interface
- Built an asynchronous parse, chunk, embed, index, and search pipeline for PDFs up to 20 MB, generating 768-dimensional embeddings for semantic retrieval
- Offloaded processing to a Redis worker using a BRPOP queue; returned HTTP 202 Accepted while tracking job status with hashes (HSET)
- Reduced hallucinations with a similarity threshold of ≥ 0.7 and cleaned up MongoDB records after processing failures to prevent orphans

MoonMovement — Startup Community Platform | *PostgreSQL, Prisma, Node.js, Redis, AWS* — GitHub | Live
- Built a Reddit-style startup platform featuring discussions, voting, user profiles, and startup discovery
- Designed a 3-tier backend using stateless APIs, PostgreSQL, and Redis, deployed on AWS EC2 and RDS
- Implemented Redis caching, rate limiting, and optimized Prisma queries to reduce database load
- Developed secure authentication, role-based access control, and asynchronous notification workflows

BugTrack — Full-Stack Bug Tracking System | *Node.js, Express.js, PostgreSQL, Prisma, JWT* — GitHub | Live
- Built a bug tracking platform supporting 50+ projects and 500+ issues, centralizing project, issue, and user management
- Developed modular REST APIs with Express.js, PostgreSQL, and Prisma; implemented JWT authentication and reusable middleware to secure endpoints

MIT Library Portal — Academic Resource Platform | *Node.js, Express.js* — GitHub | Live
- Built a past-year question paper repository for 10,000+ MIT Manipal students, generating 40,000 impressions within 10 days of launch

Technical Skills

Languages: Python, Java, C++, TypeScript, SQL
Software Engineering: Node.js, Express.js, REST APIs, Git, GitHub, Linux
Cloud & Data: AWS (EC2, RDS, Bedrock), PostgreSQL, MongoDB, Prisma, Redis, Docker, Nginx
AI & Machine Learning: LLM Applications, RAG, Vector Search, NLP, CNNs

Certifications

Andrew Ng's Machine Learning Specialization | Google Data Analytics Professional Certificate | AWS Cloud Practitioner Essentials